

JINSU HONG

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RESEARCH INTERESTS

Uncertainty Quantification; Conformal Prediction; Imbalanced and Ordinal Learning; Multimodal Learning; Foundation Models; Computer Vision; Scientific Machine Learning

EDUCATION

Georgia State University, Atlanta, GA, USA

Ph.D. in Computer Science

Aug. 2023 – Present

Advisor: Berkay Aydin

M.S. in Data Science and Analytics

Aug. 2021 – May 2023

Sungkyunkwan University, Suwon, South Korea

B.E. in Advanced Materials Science and Engineering

Mar. 2010 – Jan. 2017

RESEARCH EXPERIENCE

Research Assistant, Data Mining Laboratory, Georgia State University

Jan. 2022 – Present

- Conduct research on reliable machine learning with emphasis on uncertainty quantification, conformal prediction, imbalanced learning, ordinal learning, multimodal learning, and scientific machine learning.
- Develop class-conditional and ordinal conformal prediction methods for reliable prediction sets under class imbalance, including quantile-regression-based and structured calibration approaches.
- Develop multimodal deep-learning pipelines for solar flare forecasting using HMI and AIA observations, including convolutional neural networks and Vision Transformer architectures.
- Fine-tune and evaluate the Surya heliophysics foundation model and conventional deep neural networks for downstream solar-flare forecasting on high-resolution, multi-channel SDO imagery.
- Design reproducible PyTorch/PyTorch Lightning research pipelines for model training, calibration, benchmarking, and uncertainty-aware evaluation.

Researcher Intern, Frontier Development Lab (FDL.AI)

Jun. 2025 – Aug. 2025

- Contributed to the 2025 “Decoding Solar Wind Structure” challenge, focusing on fine-tuning a heliophysics foundation model for downstream space-weather analysis.
- Created machine-learning-ready solar-wind datasets through multi-source data integration, outlier removal, interpolation, and task-specific preprocessing.
- Developed data loaders and downstream head architectures, including MLP, skip-connection, and transformer-based designs, and collaborated with an interdisciplinary research team on model evaluation.

RESEARCH GRANTS & COMPUTING ALLOCATIONS

ACCESS Explore Allocation, NSF ACCESS Program, 2025–2026 — Principal Investigator

Project: “Fine-Tuning the Surya Heliophysics Foundation Model for Enhanced Solar Flare Forecasting.”

Allocation: 400,000 ACCESS Credits through an initial allocation and approved supplement for advanced computing resources.

PEER-REVIEWED PUBLICATIONS

1. **J. Hong**, D. Martin, and J. Gallego, “SDOFMv2: A Multi-Instrument Foundation Model for the Solar Dynamics Observatory with Transferable Downstream Applications,” *Solar Physics*, 2026. DOI: 10.1007/s11207-026-02740-z.
2. **J. Hong**, C. Pandey, and B. Aydin, “Uncertainty-Aware Solar Flare Regression,” 2025 IEEE International Conference on Data Mining Workshops (ICDMW), pp. 1976–1984, 2025. DOI: 10.1109/ICDMW69685.2025.00238.
3. C. Pandey, **J. Hong**, A. Ji, R. A. Angryk, and B. Aydin, “Ordinal Encoding as a Regularizer in Binary Loss for Solar Flare Prediction,” 2025 IEEE International Conference on Data Mining Workshops (ICDMW), pp. 1890–1895, 2025. DOI: 10.1109/ICDMW69685.2025.00227.
4. C. Pandey, T. Adeyeha, **J. Hong**, R. A. Angryk, and B. Aydin, “Advancing Solar Flare Prediction Using Deep Learning with Active Region Patches,” *ECML PKDD 2024, Applied Data Science Track*, pp. 50–65, 2024. DOI: 10.1007/978-3-031-70381-2_4.
5. **J. Hong**, C. Pandey, A. Ji, and B. Aydin, “An Innovative Solar Flare Metadata Collection for Space Weather Analytics,” 22nd International Conference on Machine Learning and Applications (ICMLA), pp. 408–413, 2023.
6. **J. Hong**, A. Ji, C. Pandey, and B. Aydin, “Enhancing Solar Flare Prediction with Innovative Data-Driven Labels,” 2023 IEEE 5th International Conference on Cognitive Machine Intelligence (CogMI), 2023.
7. **J. Hong**, A. Ji, C. Pandey, and B. Aydin, “Beyond Traditional Flare Forecasting: A Data-driven Labeling Approach for High-fidelity Predictions,” 25th International Conference on Big Data Analytics and Knowledge Discovery (DaWaK), *Lecture Notes in Computer Science*, pp. 380–385, 2023. DOI: 10.1007/978-3-031-39831-5_34.

PREPRINTS & RESEARCH MANUSCRIPTS

1. S. Roy, J. Schmude, R. Lal, et al., including **J. Hong**, “Surya: Foundation Model for Heliophysics,” arXiv:2508.14112, 2025.

SELECTED PRESENTATIONS

1. **J. Hong**, C. Pandey, and B. Aydin, “Enhancing Solar Flare Prediction with Integrated Multi-wavelength Imagery and Conformal Prediction,” AGU Annual Meeting, 2024.
2. **J. Hong** et al., “Evaluating the Effectiveness of Active Region/Polarity Inversion Line Segmentation with Heliophysics Foundation Models,” AGU presentation / ESS Open Archive, 2025.

SELECTED RESEARCH THEMES

Uncertainty Quantification and Ordinal Learning. Develop conformal prediction methods that provide reliable uncertainty sets under class imbalance and ordered label structures, with particular attention to class-conditional coverage and rare classes.

Multimodal Learning for Scientific Data. Study single- and multi-channel learning from heterogeneous solar observations, including magnetic-field and EUV imagery, using CNN and Vision Transformer architectures.

Foundation Models for Heliophysics. Adapt and evaluate large pretrained models such as Surya for downstream space-weather forecasting and related scientific tasks.

RESEARCH SOFTWARE & OPEN-SOURCE PROJECTS

Ordinal CQR. Class-conditional conformal prediction framework for ordinal outcomes combining quantile regression, Mondrian calibration, candidate-specific inference, and contiguous prediction sets. [GitHub](#)

Ordinal-Aware Conformal. Research codebase for ordinal-aware conformal prediction, structured calibration, rare-class analysis, and benchmark experiments across synthetic and real-world datasets. [GitHub](#)

Flare Surya. Foundation-model adaptation and evaluation pipeline for solar-flare forecasting using SuryaBench and high-resolution multi-channel solar observations. [GitHub](#)

Multi-Imagery Solar Flare Prediction. Multimodal deep-learning pipeline for single- and multi-channel HMI/AIA solar imagery, model comparison, calibration, and flare forecasting experiments. [GitHub](#)

PROFESSIONAL SERVICE

- Reviewer, International Conference on Machine Learning and Applications (ICMLA), 2024.
- Reviewer, International Conference on Machine Learning and Applications (ICMLA), 2023.

INDUSTRY EXPERIENCE

Metrology Engineer, SK hynix

Jan. 2017 – Jul. 2021

Chungju, South Korea

- Worked in semiconductor manufacturing as a metrology engineer before transitioning to graduate study in data science and machine learning.

TECHNICAL SKILLS

Programming: Python, C++, SQL, R

Machine Learning: PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, NumPy, Pandas, Dask

Methods: Conformal prediction, quantile regression, calibration, imbalanced learning, ordinal learning, computer vision, multimodal deep learning, foundation-model adaptation

Tools: Linux, Docker, Git/GitHub, Hydra, Matplotlib, LaTeX, MySQL, PostgreSQL

RESEARCH PROFILES

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